

Flight Readiness Review — Payload Thermal Analysis

High-Altitude Balloon Event-Camera Payload
Operating concept: on-board computer powered at 16 km

Thermal Analysis — Systems Engineering

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Abstract

This report documents the thermal model and flight-readiness assessment of the high-altitude balloon (HAB) event-camera payload for the operating concept in which the on-board computer (OBC) is powered only on reaching the imaging altitude of 16 km. A coupled three-block model (external environment, enclosure-wall conduction, and the two-dimensional axisymmetric internal field) is solved over the ascent-to-apogee mission window from a cold-night initial temperature of $+5^{\circ}\text{C}$. All seventeen payload components remain within their operating temperature limits for the full analysed window, with a minimum cold-side operating margin of $+7.6^{\circ}\text{C}$. The thermal design is assessed **GO** for flight.

Contents

1	System Description and Geometry	1
2	Modelling Methodology	2
2.1	Block A — External Environment and Forced Convection	2
2.2	Block B — One-Dimensional Wall Conduction	3
2.3	Block C — Two-Dimensional Axisymmetric Internal Field	3
2.4	Coupling	4
3	Component Database	4
4	Mission Profile	6
5	Results	6
6	Conclusion	8

1 System Description and Geometry

The payload is housed in a cylindrical expanded-polypropylene (EPP) Thermobox enclosure. The internal cavity is modelled as an axisymmetric cylinder of internal radius R_{int} and internal height L_{int} , surrounded by a uniform EPP wall of thickness L_{wall} (side and end caps). The geometry is summarised in Table 1.

Table 1: Enclosure geometry.

Symbol	Quantity	Value
R_{int}	Internal radius	0.080 m
L_{int}	Internal height	0.250 m
L_{wall}	Side-wall thickness (EPP)	0.030 m
L_{cap}	End-cap thickness (EPP)	0.030 m
R_{box}	Outer radius	0.110 m
L_{box}	Outer height	0.310 m
D_{box}	Outer diameter (convection)	0.220 m

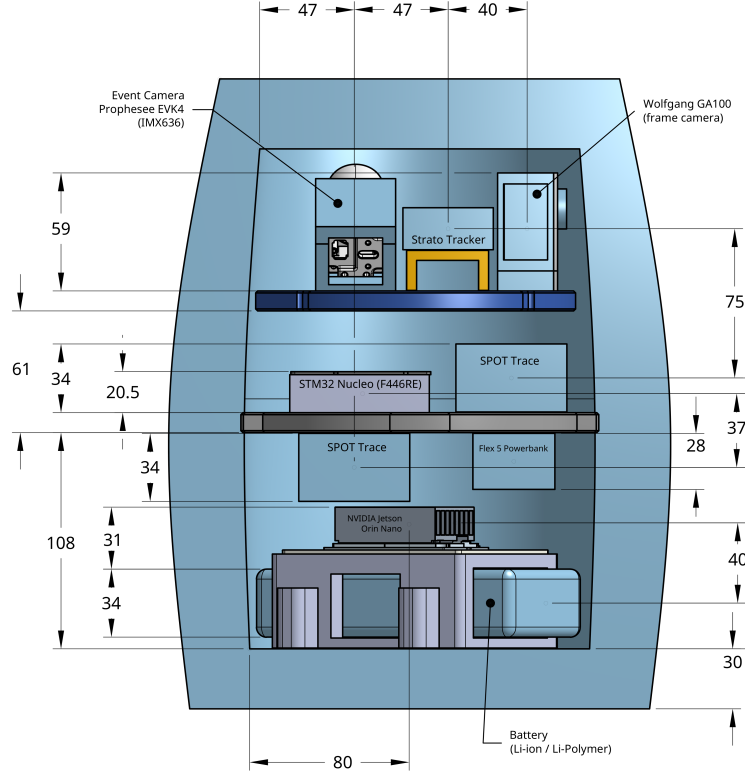


Figure 1: Mechanical CAD section used as the reference layout for the meshed model. The interior resolves four stacked rows: cameras and tracker on top (EVK4 + lens, Strato Tracker, GA100), the avionics row (STM32, SPOT Trace), the power/recovery row (GNSS/SD cluster, Flex 5 powerbank), and the heavy bottom row (Jetson Orin Nano inboard, battery outboard, on-board storage). Components are mounted with a clearance to the side wall.

2 Modelling Methodology

The model is decomposed into three coupled blocks. Block A provides the external free-stream temperature and convective coefficient as a function of altitude and ascent velocity. Block B solves transient conduction through the enclosure wall and end caps. Block C solves the two-dimensional axisymmetric temperature field of the internal cavity, driven by the component heat

loads and bounded by the Block B wall temperatures. The blocks are advanced with a lagged coupling each timestep.

2.1 Block A — External Environment and Forced Convection

The ambient state is evaluated with the US Standard Atmosphere 1976, piecewise by layer. Within a layer of base altitude h_b , base temperature T_b , base pressure P_b and lapse rate L_b ,

$$T(h) = T_b + L_b (h - h_b), \quad (1)$$

$$P(h) = \begin{cases} P_b \left(\frac{T}{T_b} \right)^{-g/(L_b R)}, & L_b \neq 0, \\ P_b \exp\left(-\frac{g(h - h_b)}{R T_b}\right), & L_b = 0, \end{cases} \quad \rho = \frac{P}{R T}, \quad (2)$$

with $g = 9.80665 \text{ m/s}^2$ and $R = 287.058 \text{ J/(kg K)}$. Transport properties use Sutherland-type fits,

$$\mu(T) = \frac{1.458 \times 10^{-6} T^{3/2}}{T + 110.4}, \quad k_{air}(T) = \frac{2.495 \times 10^{-3} T^{0.8646}}{1 + 245.4 \cdot 10^{-12/T/T}}, \quad \text{Pr} = \frac{\mu c_p}{k_{air}}. \quad (3)$$

The enclosure is treated as a cylinder in cross-flow at the ascent velocity V . The Reynolds number and the Churchill–Bernstein Nusselt-number correlation give the external convective coefficient:

$$\text{Re} = \frac{\rho V D_{box}}{\mu}, \quad (4)$$

$$\text{Nu} = 0.3 + \frac{0.62 \text{Re}^{1/2} \text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282\,000} \right)^{5/8} \right]^{4/5}, \quad h_{ext} = \frac{\text{Nu} k_{air}}{D_{box}}. \quad (5)$$

2.2 Block B — One-Dimensional Wall Conduction

Each wall (side, top cap, bottom cap) is discretised into N nodes across the thickness, with spacing $\Delta x = L_{wall}/(N - 1)$. Transient conduction is advanced with implicit Euler in a finite-volume form. Boundary nodes carry half control volumes; interior nodes carry full control volumes. The outer surface (node 0) exchanges with the atmosphere through h_{ext} and the inner surface (node $N - 1$) with the cavity air through the internal coefficient h_{int} :

$$\text{Outer: } \frac{\rho c_p \Delta x}{\Delta t} (T_0^{n+1} - T_0^n) = h_{ext} (T_\infty - T_0^{n+1}) + \frac{k}{\Delta x} (T_1^{n+1} - T_0^{n+1}), \quad (6)$$

$$\text{Interior: } \frac{\rho c_p \Delta x}{\Delta t} (T_i^{n+1} - T_i^n) = \frac{k}{\Delta x} (T_{i-1}^{n+1} - 2T_i^{n+1} + T_{i+1}^{n+1}), \quad (7)$$

$$\text{Inner: } \frac{\rho c_p \Delta x}{\Delta t} (T_{N-1}^{n+1} - T_{N-1}^n) = \frac{k}{\Delta x} (T_{N-2}^{n+1} - T_{N-1}^{n+1}) + h_{int} (T_{air} - T_{N-1}^{n+1}). \quad (8)$$

These equations form a tridiagonal system solved directly each timestep. The camera optical windows are modelled as a thin glass pane solved with the same one-dimensional scheme and coupled to the internal field as a parallel conductance on the upper ring.

The internal coefficient h_{int} represents natural convection of the cavity air. In the laminar enclosure regime the Nusselt number scales as $\text{Nu} \propto \text{Ra}^{1/4}$, and since $\text{Ra} \propto \rho^2 \propto P^2$ the coefficient varies with the square root of the local ambient pressure $P(h)$ from Block A:

$$h_{int}(h) = h_{int,0} \sqrt{\frac{P(h)}{P_0}}, \quad h_{int,0} = 1.5 \text{ W/(m}^2\text{K)}, \quad P_0 = 101,325 \text{ Pa}. \quad (9)$$

As the payload ascends and the atmosphere thins, the internal convective coupling weakens accordingly, decreasing from $1.5 \text{ W/(m}^2\text{K)}$ at the surface to approximately $0.14 \text{ W/(m}^2\text{K)}$ near apogee.

2.3 Block C — Two-Dimensional Axisymmetric Internal Field

The cavity is meshed on a uniform (r, z) grid with cell centres at $r_i = (i + \frac{1}{2})\Delta r$ and $z_j = (j + \frac{1}{2})\Delta z$. For cell i the radial faces lie at $r_i^- = i\Delta r$ and $r_i^+ = (i + 1)\Delta r$, giving the face areas and cell volume

$$A_r^- = 2\pi r_i^- \Delta z, \quad A_r^+ = 2\pi r_i^+ \Delta z, \quad A_z = \pi(r_i^{+2} - r_i^{-2}), \quad V = A_z \Delta z. \quad (10)$$

Each cell P is advanced with an implicit finite-volume energy balance over its faces f to neighbours F , with an internal source Q_P (the component dissipation assigned to the cell):

$$\frac{\rho_P c_{p,P} V_P}{\Delta t} (T_P^{n+1} - T_P^n) = \sum_f G_f (T_F^{n+1} - T_P^{n+1}) + Q_P. \quad (11)$$

Interface conductances use the harmonic-mean conductivity to handle dissimilar materials,

$$G_f = \frac{k_{harm} A_f}{d}, \quad k_{harm} = \frac{2k_P k_F}{k_P + k_F}, \quad (12)$$

where $d = \Delta r$ for radial faces and $d = \Delta z$ for axial faces. Dirichlet boundaries (the inner wall ring $T_{r,wall}(z)$ from the side wall, and the cap temperatures T_{top} , T_{bot}) use a half-cell distance $d/2$, contributing $G_{bc} = k A_f / (\Delta r/2)$ to the diagonal and $G_{bc} T_{bc}$ to the right-hand side. The axis ($i = 0$) is a symmetry boundary with zero radial flux ($A_r^- = 0$). On the upper ring the glass window adds a parallel path of area A_{win} to T_{window} alongside the wall area A_{eps} to $T_{r,wall}$. Assembling all cells yields a sparse linear system $\mathbf{A} \mathbf{T}^{n+1} = \mathbf{b}$, solved with a sparse direct solver each timestep.

2.4 Coupling

Per timestep n : (i) Block A returns T_∞ , h_{ext} and the ambient pressure from the altitude and velocity profiles; (ii) the cavity air temperature T_{air} is taken as the volume-weighted mean of the air cells and h_{int} is evaluated at the current altitude; (iii) Block B advances the side wall, caps and window pane; (iv) Block C advances the internal field using the updated inner-surface temperatures. The numerical parameters are listed in Table 2.

Table 2: Numerical and boundary parameters.

Symbol	Quantity	Value
$N_r \times N_z$	Internal mesh	32×100 (3200 cells)
$\Delta r, \Delta z$	Cell size	2.5 mm
N	Wall nodes (Block B)	5
Δt	Timestep (implicit)	30 s
t_{end}	Analysis window	6800 s (113.3 min)
$h_{int,0}$	Internal coefficient (sea level)	1.5 W/(m ² K)
T_{launch}	Initial temperature	278.15 K (5 °C)

3 Component Database

The payload comprises seventeen entries (Table 3). Each is mapped to a rectangular cell region of the mesh per the CAD layout (Fig. 1), assigned an effective material, a steady electrical dissipation, and operating temperature limits. Storage devices (SSD, USB) dissipate their drawn power as heat.

Table 3: Payload component database: material, mass, steady dissipation, and operating temperature limits.

Component	Subsys.	m [g]	P [W]	$T_{op,min}$	$T_{op,max}$
NVIDIA Jetson Orin Nano	OBC	260	7.00	-25°C	90°C
Battery (Li-ion / Li-Po)	EPS	1000	0.30	-20°C	60°C
250 GB SSD (NVMe M.2)	OBC	10	3.00	-40°C	85°C
USB Stick (USB 3.0)	OBC	5	3.00	-25°C	85°C
STM32 Nucleo F446RE	OBC	50	0.30	-40°C	85°C
GNSS Module (u-blox NEO-M9N)	Sensors	10	0.15	-40°C	85°C
SD Card Reader + Card	OBC	5	0.05	-25°C	85°C
IMU (TDK ICM-20948) $\times 2$	Sensors	5	0.01	-40°C	85°C
Env. Sensor MS8607 $\times 2$	Sensors	2	0.005	-40°C	85°C
SPOT Trace	Recovery	70	0.30	-40°C	60°C
Strato Tracker	Recovery	60	0.50	-40°C	85°C
Flex 5 Powerbank	EPS	400	0.40	-10°C	45°C
Wolfgang GA100 (frame camera)	Payload	120	3.00	-10°C	40°C
Prophesee EVK4 (IMX636)	Payload	130	1.50	-20°C	60°C
Soyo C-Mount Lens	Payload	50	0.00	-40°C	85°C
GNSS Antenna (ceramic patch)	Sensors	20	0.00	-40°C	85°C
Wiring & Connectors	EPS	200	0.50	-55°C	125°C

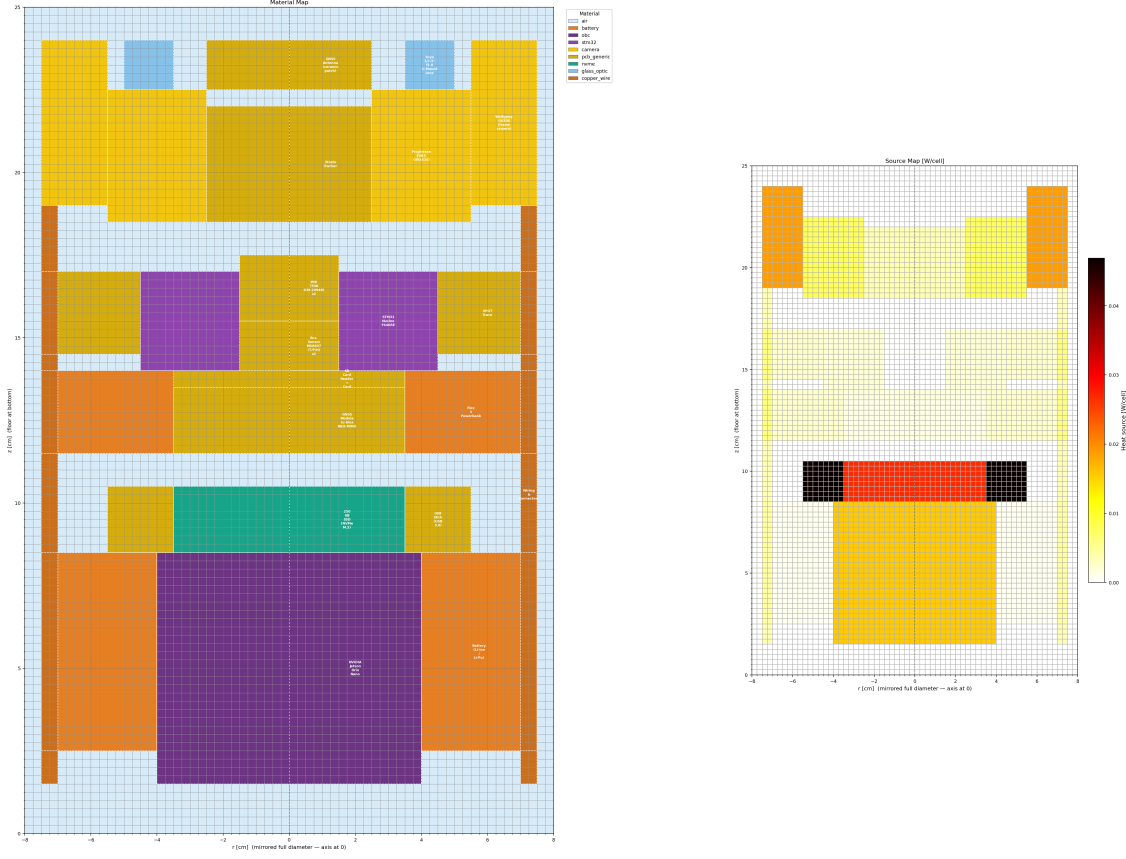


Figure 2: Compiled materials map (left) and heat-source map (right), shown in CAD orientation (z vertical, full mirrored diameter). The map reproduces the four-row layout of Fig. 1; the source map shows the Jetson as the dominant internal heat source in the bottom row.

4 Mission Profile

The analysed window is the powered ascent from launch to apogee and the first part of descent. The balloon ascends at 5 m/s to a burst altitude of 33 km ($t = 6600$ s), then descends under parachute at 5.5 m/s. The imaging altitude of 16 km is reached at $t \approx 3200$ s (53.5 min). In this operating concept the OBC (Jetson Orin Nano, 7 W) is held unpowered during ascent and is energised on reaching 16 km, remaining active for the one-hour imaging window to the end of the analysis ($t = 6800$ s); all other payload loads are active from launch. The resulting free-stream temperature and external convective coefficient are shown in Fig. 3: the ambient reaches the tropopause minimum near -56.5°C , while h_{ext} falls monotonically with the thinning atmosphere.

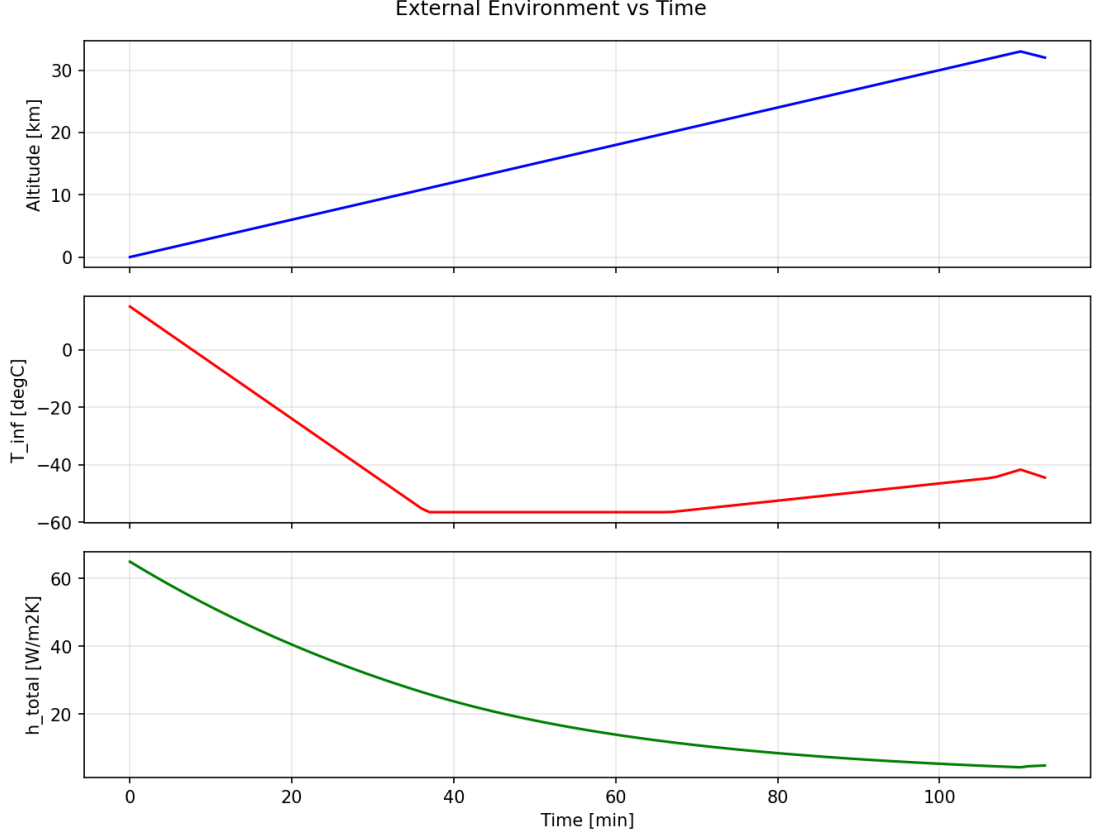


Figure 3: External environment over the mission window: altitude, free-stream temperature T_{∞} , and external convective coefficient h_{ext} from Block A.

5 Results

The internal temperature field is shown at five instants in Fig. 4. The cavity cools through the ascent with no internal heat source until the OBC is energised at 16 km, after which the Jetson establishes a warm core in the bottom row. Component temperature histories are given in Fig. 5, and the minima/maxima with cold-side operating margins are tabulated in Table 4.

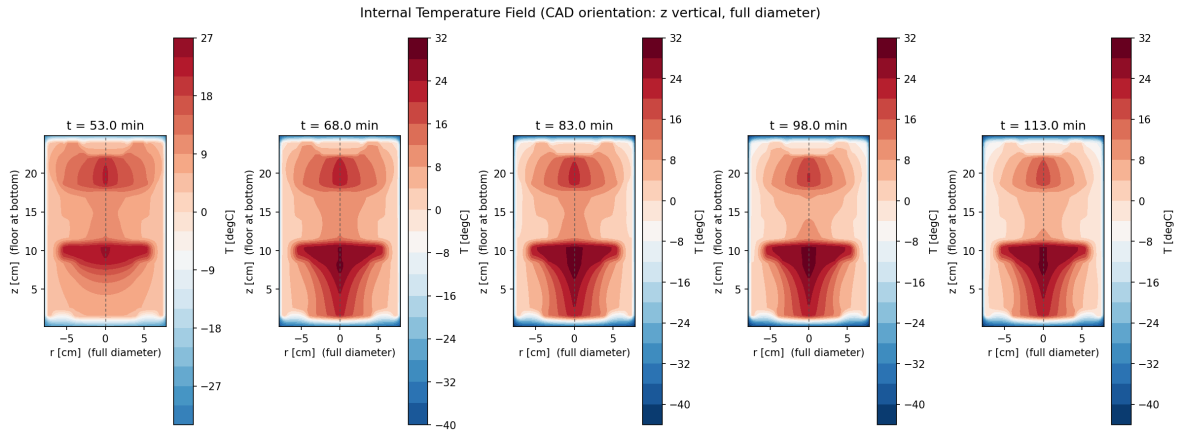


Figure 4: Internal temperature field (CAD orientation, full diameter) at five mission times. With the OBC unpowered during ascent the cavity tracks the cooling wall; the warm bottom-row core appears only after the OBC is energised at 16 km.

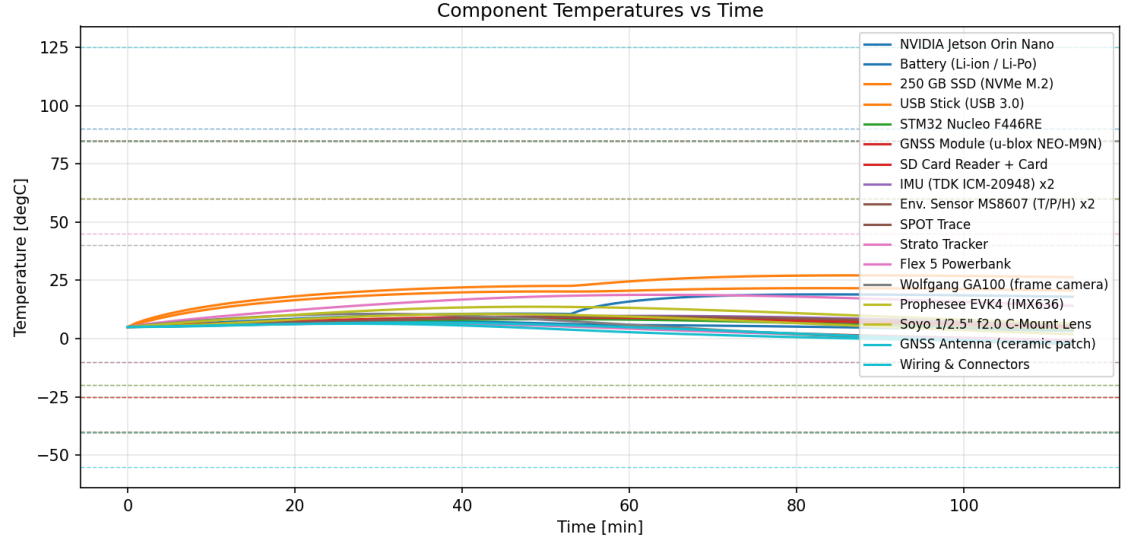


Figure 5: Component mean-temperature histories over the mission window with operating limit bands.

Table 4: Component temperature results over the analysis window and cold-side operating margin $\Delta T = T_{min} - T_{op,min}$. All components are within limits.

Component	T_{min}	T_{max}	$T_{op,min}$	ΔT	Status
NVIDIA Jetson Orin Nano	5.0	19.0	-25	+30.0	OK
Battery (Li-ion / Li-Po)	3.4	7.4	-20	+23.4	OK
250 GB SSD (NVMe M.2)	5.0	27.1	-40	+45.0	OK
USB Stick (USB 3.0)	5.0	21.7	-25	+30.0	OK
STM32 Nucleo F446RE	3.8	8.3	-40	+43.8	OK
GNSS Module (u-blox NEO-M9N)	5.0	9.5	-40	+45.0	OK
SD Card Reader + Card	4.8	9.0	-25	+29.8	OK
IMU (TDK ICM-20948) $\times 2$	5.0	9.8	-40	+45.0	OK
Env. Sensor MS8607 $\times 2$	5.0	9.4	-40	+45.0	OK
SPOT Trace	-0.7	6.7	-40	+39.3	OK
Strato Tracker	5.0	18.8	-40	+45.0	OK
Flex 5 Powerbank	-0.8	6.6	-10	+9.2	OK
Wolfgang GA100 (frame camera)	-2.4	10.6	-10	+7.6	OK
Prophesee EVK4 (IMX636)	5.0	13.7	-20	+25.0	OK
Soyo C-Mount Lens	1.9	10.6	-40	+41.9	OK
GNSS Antenna (ceramic patch)	-2.1	6.9	-40	+37.9	OK
Wiring & Connectors	-1.6	6.4	-55	+53.4	OK

All temperatures are in $^{\circ}\text{C}$. The minimum operating margin across the payload is $+7.6^{\circ}\text{C}$ on the Wolfgang GA100 frame camera, the most cold-limited unit ($T_{op,min} = -10^{\circ}\text{C}$), which is mounted in the top-outer corner and is therefore the most exposed to the cold side wall and top cap. The Flex 5 powerbank is the second most constrained unit at $+9.2^{\circ}\text{C}$. On the hot side, the warmest component is the SSD at 27.1°C against an 85°C limit and the Jetson peaks at 19.0°C against 90°C ; hot-side margins therefore exceed 50°C throughout.

6 Conclusion

The coupled three-block thermal model, built directly on the mechanical CAD layout, shows that with the OBC powered at 16 km all seventeen payload components remain within their operating temperature limits throughout the powered ascent-to-apogee window from a $+5^{\circ}\text{C}$ initial temperature, with a minimum cold-side margin of $+7.6^{\circ}\text{C}$. On this basis the payload thermal design is assessed **GO for flight**.